IEEE CASE 2025 Hackathon Challenge Problem:

Image-based Laser Powder Bed Fusion Additive Manufacturing Process Monitoring

Background

Laser Powder Bed Fusion (LPBF) is a leading metal additive manufacturing (AM) technology valued for its ability to fabricate complex, high-performance components. However, the process remains vulnerable to faults that compromise process stability and part quality. One of the most critical—and often overlooked—process faults is associated with powder spreading, where anomalies like streaks or debris can happen and lead to uneven powder layers and downstream abnormal thermal behaviors. Monitoring the layer-wise powder spreading conditions is essential to detect potential defects early in the printing process. As machine vision and data-driven methods become more capable, image-based inspection has become a promising tool to automate the quality control of powder spreading process. Still, challenges remain in robustly detecting anomalies under varying lighting and surface conditions, and in generating high-fidelity synthetic data to support model development.

Objective

This challenge invites participants to explore two problem tasks: building models to segment powder spreading anomalies and generating realistic powder-bed images to enhance quality monitoring in LPBF.

Dataset Set Description

This dataset was collected when printing a batch of four identical parts with overhang features at National Institute of Standards and Technology (NIST). The specific data used for this challenge are from one of the two process monitoring systems- the layer camera, capturing images before and after the laser scan at each layer, and under three different lighting conditions. The details of part design, fabrication and process monitoring can be found in the reference [1].

For each layer between the 2nd and the 250th layer, layer-wise images were captured before scan (A) to inspect the powder spreading quality and after scan (B) to characterize the print quality of this layer (Figure 1). To reveal various types of anomalies and defects, layer-wise images were captured with three lighting conditions (a, b, and c) individually activated. The regions of interest (ROIs) corresponding to the scan areas of each of the four parts were cropped from the layer-wise images, registered and named according to the nomenclature illustrated in Figure 1. Note that the ROI has a size of 139×250 pixels. The entire dataset has 5,976 cropped images in total (249 layers × 2 (before &after exposure) x 3 lighting conditions × 4 parts).

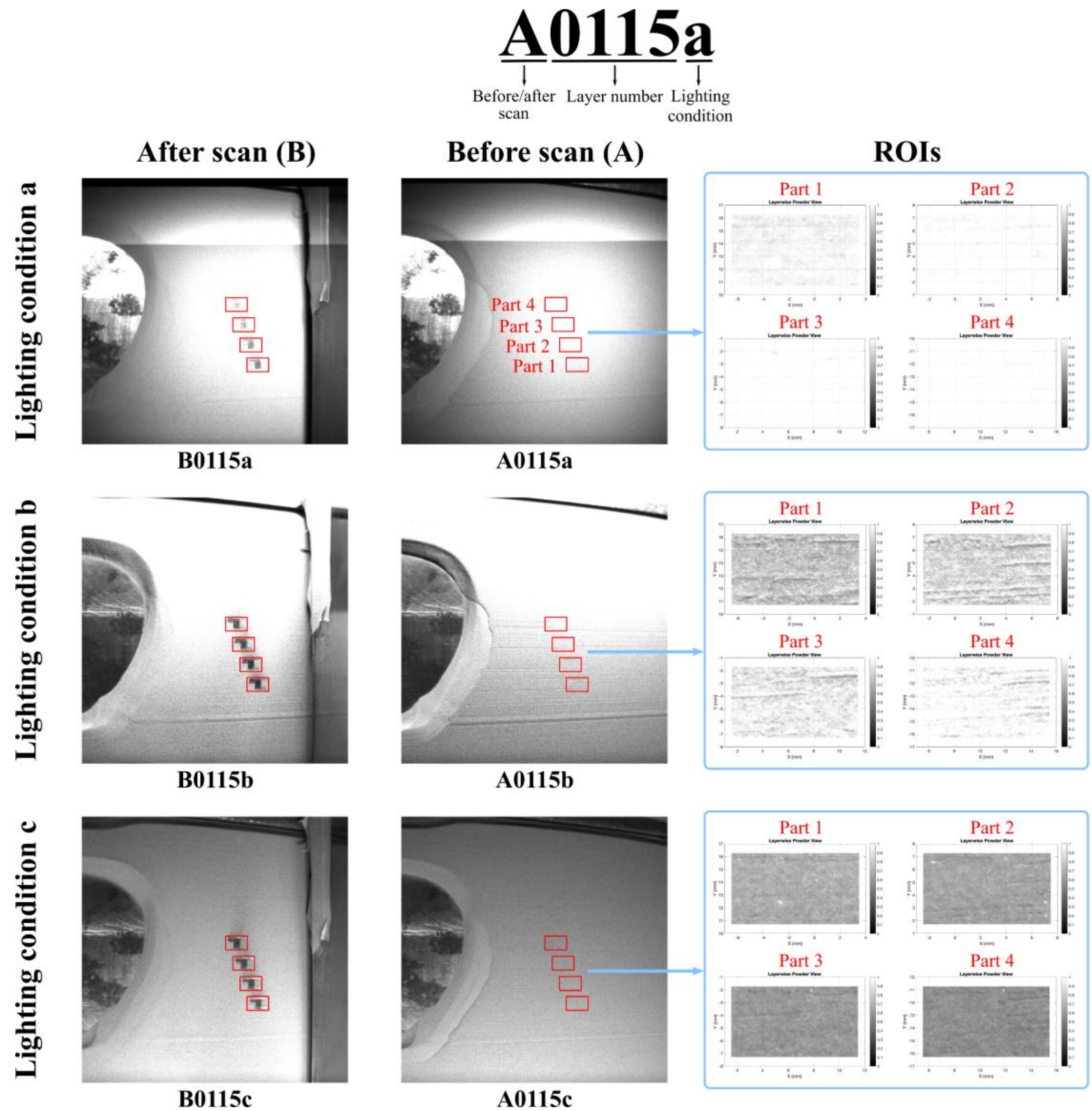
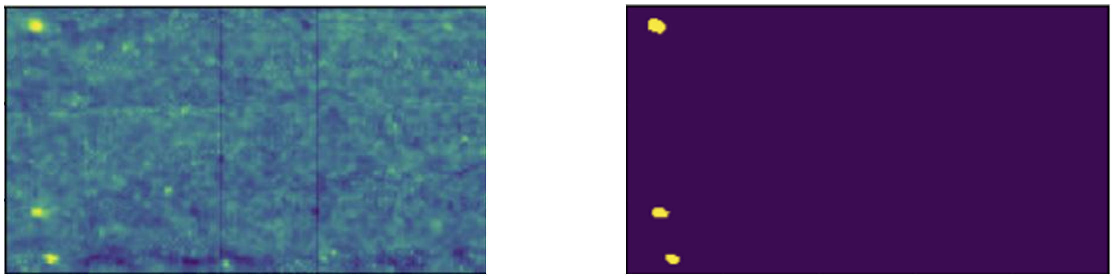


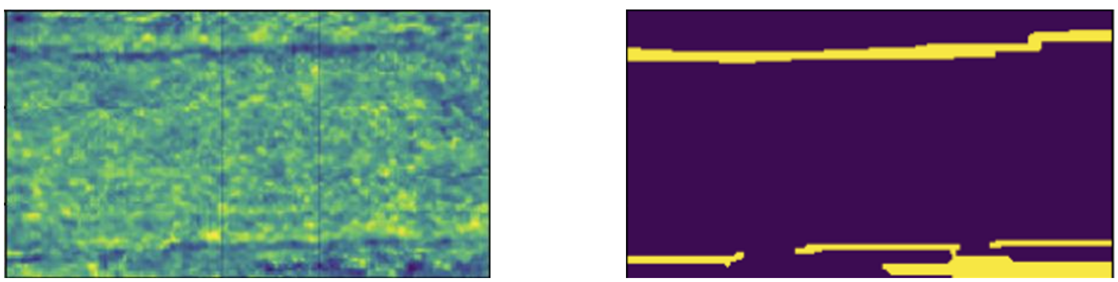
Figure 1: Nomenclature and data registration of the dataset.

In this NIST challenge, participants are provided with the [registered ROIs extracted from the images before laser scanning] (Axxxxa, Axxxxb and Axxxxc), which reflects the powder spreading quality. We focus on the two most frequently observed powder spreading anomalies in this dataset: bright spots and streaks. Bright spots usually occur at the edge of a part, signifying an uneven height along the edge. When the edge is higher than the designated height, the metal powder above will be thinner, making the part poorly covered. This leads to reflections and thus bright spots under certain lighting conditions. Figure 2 presents an example of a preprocessed ROI and its associated bright spot anomaly mask. We preprocessed the original ROI for better visualization (Figure 2a). The associated positive mask indicates the regions where bright spots were identified from the ROI (Figure 2b). The positive mask of each ROI is converted to a binary NumPy array of size 139 × 250 pixels, with an element = 0 indicating a non-bright spot and an element = 1 indicating a bright spot. Similarly, Figure 3 presents an example of a preprocessed ROI and its associated streak anomaly mask. The associated positive mask indicates the regions where streaks were identified from the ROI (Figure 3b).



(a) **(b)**

Figure 2: Example of the bright spot anomaly: a) Preprocessed ROI and b) Bright spot positive mask.



(a) **(b)**

Figure 3: Example of the streak anomaly: a) Preprocessed ROI and b) Streak positive mask.

In this competition, we randomly select 996 ROIs from the four parts and split them into a labeled training set, an unlabeled training set, and a test set at a ratio of 2:1:1 (Table 1). The labeled training set (labeled_training_set.pkl) contains the ROIs from 498 layers, along with the associated bright spot and streak positive masks for supervised learning. The unlabeled training set (unlabeled_training_set.pkl) contains the layer-wise images from 249 layers with no associated positive masks. It is up to the participant whether and how they use unlabeled_training_set.pkl. The test set (test_set.pkl) contains the layer-wise images from 249 layers with no associated positive masks to test the model(s). The datasets can be accessed here: [Link].

Table 1: Dataset overview

Dataset	Filename	Number of layers	With anomaly masks?
Labeled training set	labeled_training_set.pkl	498	Yes
Unlabeled training set	unlabeled_training_set.pkl	249	No
Test set	test_set.pkl	249	No

labeled_training_set.pkl is a Python dictionary with a structure shown in Algorithm 1. 498 randomly selected layers from the four parts are populated in this dictionary. For each layer, the ROIs under three lighting conditions and the two positive masks are stored as NumPy arrays. Some example layers retrieved from this dictionary are shown in Figure 4. unlabeled_training_set.pkl and test_set.pkl follow a similar structure but have 'spot_label' and 'streak_label' values of None.

Algorithm 1. Structure of the Python dictionary named labeled_training_set.pkl

```
labeled_training_set.pkl{
        'part04': [
                {

                'layer_id': '0093',

                'images': {
                                'A0093a': ROI under light condition a,
                                'A0093b': ROI under light condition b,
                                'A0093c': ROI under light condition c
                        },
                'spot_label': bright spot positive mask,
                'streak_label': streak positive mask
                },
                {
                'layer_id': '0146',
                …
                },
                {
                'layer_id': '0057',
                …
                },
                …
        ],
        'part03': […],
        'part02': […],
        'part01': […]
}
```

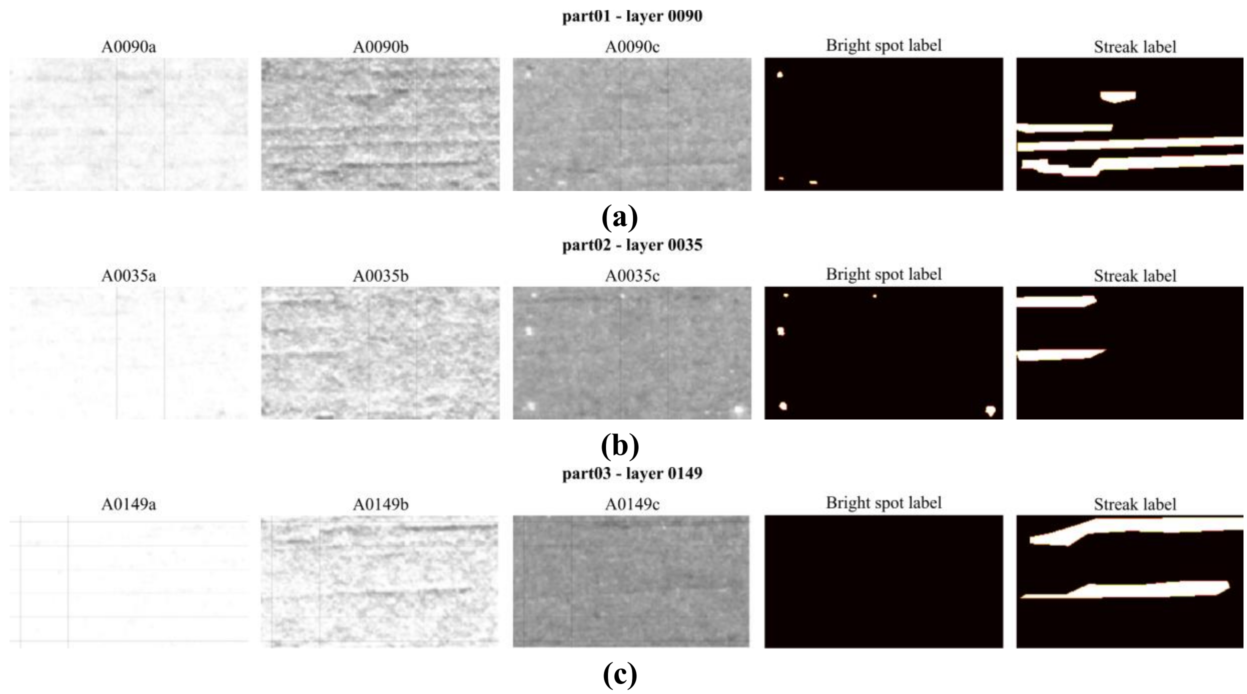


Figure 4: Examples retrieved from the dictionary.

Challenge Tasks

Task 1 – Anomaly Segmentation for Powder Spreading Monitoring

Participants will develop ML models to segment powder spreading anomalies (bright spots and streaks) from the provided ROIs, using images captured under three lighting conditions (a, b, and c) as optional model inputs (Figure 5). The models must output two separate binary masks per layer (139×250 pixels each), one for streaks and one for bright spots. For evaluation, these generated masks must be pasted to each layer in the test set dictionary as their ‘spot_label’ and ‘streak_label’, similar to the structure of the labeled training set (Algorithm 1). Please name your submission for Task 1 as ‘NIST_Task1.pkl’.

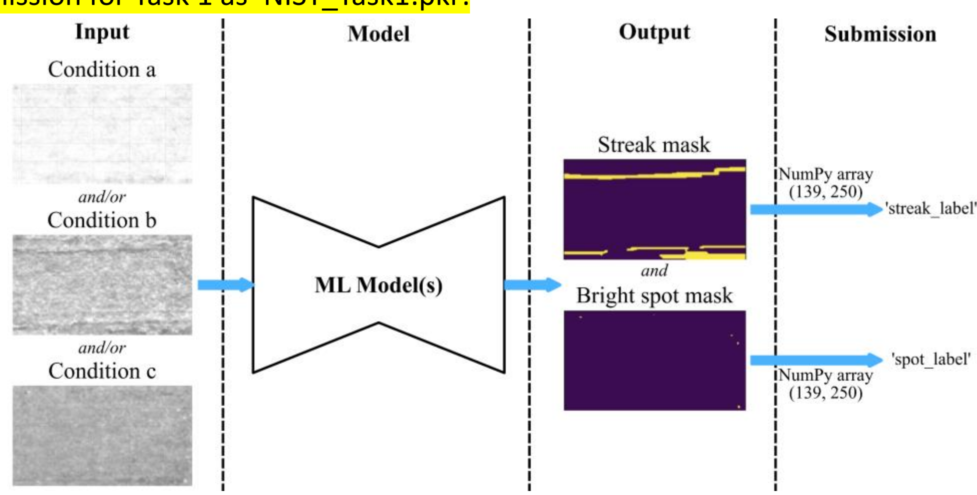


Figure 5: Pipeline of Task 1.

Submissions will be evaluated using the Intersection over Union (IoU) metric, which measures the spatial overlap between predicted anomaly masks and ground truth labels. For each anomaly type (bright spots and streaks), the IoU will be calculated separately by comparing the binary prediction masks (value=1 for anomalies) against the corresponding ground truth masks across all layers. The final score for each submission will be computed as the mean IoU averaged across both anomaly types and all test layers (249 total), providing a comprehensive assessment of segmentation accuracy while equally weighting both types of powder-spreading defects.

Task 2 – Generation of Synthetic Powder Spreading Images

Using the labeled and unlabeled training sets, participants will train a generative model to generate 100 synthetic powder spreading ROIs for each lighting condition. Please name the files as 'NIST_Task2_a.pkl', 'NIST_Task2_b.pkl', and 'NIST_Task2_c.pkl' for your synthetic ROIs for lighting conditions a, b, and c, respectively. The synthetic samples will be evaluated based on their fidelity and diversity. Synthetic samples with high fidelity should resemble real powder spreading images. For each lighting condition, the Fréchet Inception Distance (FID) between synthetic and real ROIs will be computed to assess the fidelity. First, both real and synthetic images are resized and fed into a pre-trained Inception-v3 neural network to extract their key features. Next, we calculate two things for each set of images: the average of these features (mean vector) and how they vary (covariance matrix). The FID score is then computed by combining how different these averages are and how different their variations are. A lower score means your synthetic images closely match the real ones.

To assess the diversity of your synthetic powder-spreading ROIs, we will compute the Learned Perceptual Image Patch Similarity (LPIPS) between randomly sampled pairs of your generated images (Intra-LPIPS). Intra-LPIPS evaluates diversity by comparing how differently your synthetic images "look" to a deep neural network (a pre-trained VGG). Unlike pixel-level metrics, it uses the network's learned features to mimic human perception of differences. For each pair of synthetic images, we:

1. Pass both through VGG to extract hierarchical visual features (edges, textures, patterns).
2. Compute weighted L2 distances between these features across network layers.
3. Average these distances across 100+ random image pairs in your submission.

Submission and Evaluation

You will need to submit the following items for model performance evaluation, which account for 45% of the final score:

1. NIST_Task1.pkl: A dictionary storing the anomaly segmentation masks of the test set. Please paste the generated masks to the test set dictionary such that the resulting dictionary structure resembles Algorithm 1.

2. NIST_Task2_a.pkl: A 3-D NumPy array of shape (100,139,250,3) storing the synthetic powder spreading ROIs for lighting condition a.
3. NIST_Task2_b.pkl: A 3-D NumPy array of shape (100,139,250,3) storing the synthetic powder spreading ROIs for lighting condition b.
4. NIST_Task2_c.pkl: A 3-D NumPy array of shape (100,139,250,3) storing the synthetic powder spreading ROIs for lighting condition c.

Table 2: Rubric of the challenge.

Category	Criteria	Scoring
Technical Approach (35%)	• Problem formulation • Literature review and exploration of ideas • Development and design of the idea • Scientific soundness of the approach • Creativity of the approach • Readiness of the idea and approach	Excellent (9-10 pts) Very good (7-8 pts) Good (5-6 pts) Limited (3-4 pts) Poor (1-2 pts)
Results (45%)	• Task 1: 50% • Task 2 – Fidelity: 25% • Task 2 – Diversity: 25%	Excellent (9-10 pts) Very good (7-8 pts) Good (5-6 pts) Limited (3-4 pts) Poor (1-2 pts)
Overall Presentation (20%)	• Title, headings, labels: appropriate size, location, spelling, and content • Structure and clarity	Excellent (9-10 pts) Very good (7-8 pts) Good (5-6 pts) Limited (3-4 pts) Poor (1-2 pts)

Reference:

1. Lane, B., & Yeung, H. (2020). Process monitoring dataset from the additive manufacturing metrology testbed (AMMT): Overhang part x4. Journal of research of the National Institute of Standards and Technology, 125, 1-18. https://nvlpubs.nist.gov/nistpubs/jres/125/jres.125.027.pdf